YANG-YANG LI

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Educational Background

SBU, Ph.D. in Physics, Department of Physics and Astronomy 09/2025–Present
Advisor: Tzu-Chieh Wei
Research interests: Tensor networks; symmetry and topological phases; conformal field theory; fuzzy sphere; Hopf algebra.

USTC, B.S. in Physics, Department of Modern Physics 09/2021–07/2025
School of Physical Sciences; Major: Theoretical Physics
GPA: 4.08/4.30; Rank: 4/150, 1/48

Experiences

Uniform Spectral Gaps in Two-Dimensional AKLT Models with Random Bell Bonds and Spin-1 Decorations

Doctoral Researcher, SBU Ongoing; manuscript in preparation

Advisor: Tzu-Chieh Wei **Collaborators:** Nicholas Pomata, Wenhan Guo

- Lead the analytical and numerical work on **system-size- and configuration-independent gap bounds** for two-dimensional AKLT models with random Bell bonds and spin-1 decorations.
- Combine local gap criteria with tensor-network ground-space maps.
- Classify random-bond patterns using $\mathbb{Z}_2 \times \mathbb{Z}_2$ loop invariants, Clifford rotations, and spatial symmetries; obtain explicit positive bounds on the honeycomb, square-octagon, star, and cross lattices.
- Develop **projective-metric covering bounds** for random decorations; on the cross lattice, reduce 16,791,552 symmetry-reduced cases to 686 representative overlap calculations.
- Submitted an in-person oral abstract as presenting author to the 2027 APS Global Physics Summit (Abstract ID: 4550724).

C-R-T Fractionalization in First-Quantized Hamiltonian Theory

Undergraduate Researcher 07/2024–12/2024

Supervisors: Yi-Zhuang You, Juven Wang **Collaborators:** Zheyan Wan, Shing-Tung Yau

- Formulated Majorana fermions as Clifford-algebra representations and analyzed their charge-conjugation, reflection, time-reversal, and internal symmetries in all dimensions, including $d + 1 = 5, 6, 7 \pmod 8$.
- Determined **C-R-T–internal symmetry groups** for Majorana and Dirac fermions and examined how symmetries act on the manifolds of allowed fermion bilinear masses.
- Established relations between symmetry groups in different dimensions through **domain-wall reduction**; results published in *Physical Review B*.

Many-Body Lattice C-R-T Symmetry: Fractionalization, Anomaly, and Symmetric Mass Generation

Undergraduate Researcher 07/2024–12/2024

Supervisors: Yi-Zhuang You, Juven Wang

- Derived C-R-T–internal symmetry groups for lattice Majorana and Dirac fermions and analyzed infrared/ultraviolet symmetry matching, including emergent and emanant symmetries.
- Classified **symmetric mass generation in** $0 + 1$ **through** $3 + 1$ **dimensions** and generalized to higher dimensions; used anomaly analysis and Mathematica searches to construct explicit symmetry-preserving gapping interactions.
- Checked the interactions numerically using the **Jordan–Wigner transformation** and analytically through **stabilizer analysis**; results reported in arXiv:2412.19691.

Floating Edge Bands in the BHZ Model with Altermagnetism

Undergraduate Researcher 06/2024–08/2024

Supervisor: Song-Bo Zhang

- Studied floating edge bands detached from the bulk bands throughout the Brillouin zone; **derived phase diagrams analytically** using Chern and winding numbers.
- Obtained **nonperturbative edge-state spectra and wave functions** and computed spin-resolved surface densities of states using iterative Green-function methods.
- Tested robustness against geometric rotation, energy asymmetry, and spin coupling; results published in *Physical Review B*.

Publications

[1] **Y.-Y. Li**, Z. Wan, J. Wang, S.-T. Yau, and Y.-Z. You, C-R-T fractionalization in first-quantized Hamiltonian theory, *Phys. Rev. B* **113**, 195102 (2026).

[2] **Y.-Y. Li** and S.-B. Zhang, Floating edge bands in the Bernevig–Hughes–Zhang model with altermagnetism, *Phys. Rev. B* **111**, 045106 (2025).

[3] **Y.-Y. Li**, J. Wang, and Y.-Z. You, Quantum many-body lattice C-R-T symmetry: Fractionalization, anomaly, and symmetric mass generation, arXiv:2412.19691 (2024; revised 2025).

Skills

Research methods: Tensor networks; exact diagonalization; Krylov/Lanczos methods; Clifford algebras and lattice fermions; symmetry and anomaly analysis; topological invariants; surface Green functions; stabilizer methods.

Scientific computing: Julia; Python; Sagemath; GAP; Mathematica.

Awards

· Silsbee Prize, SBU 05/2026

· Guo Moruo Scholarship, USTC (35/1950) 03/2025

· National Scholarship, USTC (11/150) 11/2024

· Excellent Student Scholarship, Silver Award, USTC (top 10%) 12/2023

· 862 Alumni Excellent Freshman Scholarship, USTC (3/379) 11/2022

Academic Activities and Memberships

· Workshop on Quantum Fields, Matter and their Symmetries 11/30/2026–12/04/2026
Institute for Advanced Study, Princeton; participant (upcoming).

· European Tensor Networks School 10/26/2026–10/30/2026
Isaac Newton Institute, Cambridge; participant (upcoming).

· Princeton Summer School on Condensed Matter Physics, Participant 07/20/2026–07/24/2026

· American Physical Society, Graduate Student Member 10/2026–09/2027
Member, Forum on Graduate Student Affairs.